

10(a)	any four from: 1 changes in factor / stimulus, detected by receptor ; 2 <i>ref. to</i> CNS / brain / coordinator ; 3 impulses / (named) hormone, sent to, (named) effector / muscle / gland ; 4 (named) effector / muscle / gland, carries out response ; 5 factor returns to, set point / norm ; 6 negative feedback ; 7 AVP ; e.g. cell signalling	4
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Question	Answer	Marks
10(b)	any three from: cell signalling carries on 1 / 2 two details ;; e.g. glucagon binds to receptors / adenylyl cyclase stimulated / G protein activated / cAMP formed / protein kinase A activated / enzyme cascade 3 no / less, (functioning) glycogen phosphorylase (produced) OR non-functioning glycogen phosphorylase (produced) OR no / less, glycogen phosphorylase activated ; 4 change in, tertiary structure / active site ; 5 (so) less / no, glycogen converted to glucose / glycogenolysis ;	3
10(c)	any three from: 1 no / less, (functioning) glycogen synthase (produced) or non-functioning glycogen synthase (produced) or no / less, glycogen synthase activated ; 2 high / increase in, blood glucose <u>concentration</u> (following this meal) ; 3 (so) glucose (excreted) in urine ; 4 more lipid synthesis ; 5 inhibits release of glucagon ; 6 AVP ; e.g. affects blood water potential / dehydration / thirst / tiredness / coma / affects blood pressure	3